

Cryptocurrency Price Prediction

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Abstract

This project investigates the prediction of next-day closing prices for five major cryptocurrencies using deep learning models implemented in PyTorch. We compare a traditional additive model (Facebook Prophet) against Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks trained on different feature sets and loss functions. Specifically, we evaluate seven model configurations: Prophet as a baseline, basic LSTM and GRU using standard technical indicators, LSTM and GRU enhanced with Geometric Brownian Motion (GBM) inspired statistical features trained with MSE loss, and LSTM and GRU with GBM features trained with a custom directional penalty loss. Our results show that GBM-enhanced recurrent models consistently achieve the lowest price forecast errors, with GBM-LSTM and GBM-GRU variants reaching comparable performance. Furthermore, LSTM and GRU architectures yield near-identical results when paired with the same features and loss, suggesting that architecture choice matters far less than feature engineering. However, all models struggle to predict price direction significantly better than random chance, and the directional penalty loss does not yield a robust improvement in directional accuracy.

1 Introduction

Cryptocurrencies are digital assets that use cryptography and decentralized blockchain or distributed ledger technology to secure transactions, control issuance, and enable peer-to-peer transfers without central authorities. Since the launch of Bitcoin in 2009, the cryptocurrency market has grown into a multi-trillion-dollar asset class with thousands of tradable tokens. In this project, five major cryptocurrencies are studied: Bitcoin (BTC), Ethereum (ETH), BNB, Solana (SOL), and XRP.

From a time-series perspective, cryptocurrency prices are particularly challenging to model. They exhibit extreme volatility, pronounced non-stationarity, unstable trends, limited or shifting seasonality, and high levels of noise. These characteristics complicate the application of traditional forecasting techniques that assume relatively stable patterns over time. At the same time, the high liquidity and 24/7 trading make cryptocurrencies a natural testing ground for modern machine learning methods.

The practical motivation of this project is to explore whether deep learning models, in particular LSTM and GRU networks implemented in PyTorch, can provide more accurate and useful forecasts than a widely used statistical benchmark model, Facebook Prophet, for short-horizon cryptocurrency prediction. An additional goal is to understand how different feature engineering strategies and loss function designs affect forecast quality. Because traders care not only about the magnitude of price changes but also about the direction of movement, the analysis considers both price forecast accuracy and directional accuracy.

The project compares seven models on the task of predicting the next-day closing price for BTC, ETH, BNB, SOL, and XRP:

- Model 1: Facebook Prophet with a 90-day rolling training window (baseline).
- Model 2: LSTM using technical indicators (RSI, SMA) and volume.
- Model 3: LSTM using GBM features (volatility and distance to SMA) with MSE loss.
- Model 4: LSTM using GBM features with a directional penalty loss.
- Model 5: GRU using technical indicators and volume.
- Model 6: GRU using GBM features with MSE loss.
- Model 7: GRU using GBM features with a directional penalty loss.

Under the experimental setup described below, several findings emerge. First, all LSTM and GRU models substantially outperform Prophet in price forecast accuracy across all five cryptocurrencies. Second, GBM-inspired statistical features are more informative than traditional technical indicators for both LSTM and GRU architectures. Third, LSTM and GRU yield near-identical performance when using the same features and loss function,

indicating that architecture choice is secondary to feature engineering. Fourth, introducing a directional penalty in the loss function does not lead to a robust improvement in directional accuracy.

2 Research Questions

The project is guided by four related research questions:

1. Among the considered models, which approach provides the most accurate next-day closing price forecasts for major cryptocurrencies: a traditional time-series model (Prophet) or recurrent neural networks (LSTM and GRU)?
2. Within the recurrent network framework, which type of features is more informative for prediction: standard technical indicators (RSI and SMA) or GBM-inspired statistical features (volatility and distance to a trend)?
3. How do LSTM and GRU architectures compare when trained on the same features and loss function?
4. Does explicitly penalizing directionally incorrect predictions in the loss function improve directional accuracy in practice?

Answering these questions helps clarify both the relative value of deep learning versus traditional models for cryptocurrency forecasting and the roles of architecture choice, input features, and training objectives.

3 Data

Source and Coverage

The dataset consists of daily OHLCV (Open, High, Low, Close, Volume) data for five cryptocurrencies traded in USD. The data were downloaded from the exchange [Bitget](#). The coverage periods are:

- Bitcoin, Ethereum, BNB, and XRP: from 2018-08-22 to 2025-11-30.
- Solana: from 2020-04-11 to 2025-11-30.

For each trading day, the raw data includes at least the opening time, the OHLC prices, and the base-asset trading volume. All Unix timestamps are converted to calendar dates and aggregated at the daily frequency.

Cryptocurrency markets trade continuously, but the downloaded daily series may contain missing calendar days due to data issues. To create a consistent daily panel, the missing days are filled using forward fill, so that any gap takes the last available observation. This yields a gap-free daily time series for each cryptocurrency.

Feature Engineering

All LSTM and GRU models predict next-day log returns, while Prophet directly models price levels. Let P_t denote the closing price on day t . The daily log return is defined as

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right).$$

This series is used both as the prediction target (for recurrent models) and as an input feature.

In addition to log returns and volumes, several derived features are constructed.

Technical indicators (Models 2 and 5). For the LSTM and GRU with technical indicators, the following quantities are used:

- A 20-day simple moving average (SMA) of the closing price,

$$\text{SMA}_{20,t} = \frac{1}{20} \sum_{k=0}^{19} P_{t-k}.$$

- A 14-day Relative Strength Index (RSI), computed using the standard Wilder smoothing method, based on average gains and losses over the prior 14 days.

GBM-inspired statistical features (Models 3, 4, 6 and 7). To capture local trend and volatility in a way that is directly motivated by the Geometric Brownian Motion (GBM) model, rolling statistics of log returns over a 90-day window are computed. Under a GBM,

$$dP_t = \mu P_t dt + \sigma P_t dW_t,$$

so the drift μ and volatility σ can be approximated in discrete time by the sample mean and standard deviation of recent log returns. For each day t :

$$\hat{\mu}_t = \frac{1}{90} \sum_{k=1}^{90} r_{t-k}, \quad \hat{\sigma}_t = \sqrt{\frac{1}{89} \sum_{k=1}^{90} (r_{t-k} - \hat{\mu}_t)^2}.$$

In the implemented models, the volatility estimate $\hat{\sigma}_t$ is used as a feature and referred to as GBM volatility.

A second feature measures how far the current price is from a short-term trend:

$$\text{dist_SMA}_t = \frac{P_t - \text{SMA}_{20,t}}{\text{SMA}_{20,t}}.$$

This relative distance to the 20-day SMA indicates whether the asset is trading above or below its recent average in percentage terms.

Handling of missing values. Computation of moving averages, RSI, and rolling GBM statistics naturally introduces missing values at the beginning of each series. After all features are created, any rows containing missing values in either inputs or targets are dropped. The remaining observations form the basis for modeling.

Train, Validation, and Test Splits

For the LSTM and GRU models, each asset’s dataset is split chronologically into training, validation, and test sets in the ratio 70%:15%:15%. No shuffling is applied, in order to avoid look-ahead bias and to reflect genuine forecasting conditions. The training set is used to fit model parameters, the validation set to tune hyperparameters and perform early stopping, and the test set to evaluate out-of-sample performance.

For Prophet, the last 15% of each asset’s observations are treated as the test period. For each test day, Prophet is re-fit using only the most recent 90 days of data and then asked to produce a one-day-ahead forecast. This rolling scheme keeps Prophet’s effective information set comparable to that of the recurrent models, which also use a 90-day lookback window.

4 Methodology

Evaluation Metrics

Model performance is evaluated along two dimensions: accuracy of price level forecasts and accuracy of direction predictions.

Let P_t be the actual closing price on test day t and $\hat{P}_t$ the corresponding model forecast. The main error metrics on price levels are Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE).

Directional accuracy (DA) measures how often the model correctly predicts whether the price will go up or down. For Prophet, which forecasts price levels directly, the actual and predicted price changes are

$$\Delta P_t = P_t - P_{t-1}, \quad \Delta \hat{P}_t = \hat{P}_t - P_{t-1}.$$

For the LSTM and GRU models, the actual and predicted log returns r_t and $\hat{r}_t$ serve the same role. In either case, directional accuracy is computed as

$$\text{DA} = \frac{100}{T} \sum_{t=1}^T 1\{\text{sign}(\Delta P_t) = \text{sign}(\Delta \hat{P}_t)\} \text{ (in percent),}$$

where $1\{\cdot\}$ is the indicator function and the sign of zero is treated neutrally. A DA of 50% corresponds to random guessing in a symmetric up/down environment, so values only

slightly above 50% should be interpreted with caution.

Model 1: Facebook Prophet

Facebook Prophet is an additive time-series model that decomposes a time series into trend, seasonality, holiday effects, and noise:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t,$$

where $g(t)$ is a trend component, $s(t)$ captures seasonality, $h(t)$ captures holiday or event effects, and ε_t is an idiosyncratic error term.

In this project, only the trend component is used, because cryptocurrencies do not exhibit stable daily, weekly, or annual seasonal patterns in the same way that retail sales or web traffic might. All built-in seasonality and holiday terms are therefore disabled. Prophet effectively fits a flexible piecewise linear trend with automatically selected change-points over the most recent 90 days of closing prices and then extrapolates one day ahead.

For each test day and each asset, Prophet is re-fit on a rolling 90-day window and outputs a one-step-ahead prediction of the next-day closing price. These rolling forecasts are then compared to realized prices over the test period to compute RMSE, MAE, MAPE, and DA.

Models 2 & 5: LSTM and GRU with Technical Indicators

Long Short-Term Memory (LSTM) networks and Gated Recurrent Unit (GRU) networks are two classes of Recurrent Neural Networks (RNN) designed to capture long-range dependencies in sequential data. LSTM networks use an explicit memory cell with input, forget, and output gates, while GRU simplifies this with reset and update gates. Both are well suited to financial time series, where the effect of past shocks may persist over many days.

Models 2 (LSTM) and 5 (GRU) predict next-day log returns based on the past 90 days of log returns, RSI, SMA, and volumes.

All features and the target variable (log return) are scaled using Min-Max scaling fitted on the training set and then applied to validation and test sets. The network architecture consists of two stacked recurrent layers with 64 units, interleaved with dropout layers ($p = 0.2$) for regularization, followed by a single linear output neuron. The loss function is the standard Mean Squared Error (MSE) between the scaled predicted and true log returns, and the model is trained with the Adam optimizer using a learning rate of 0.001. Early stopping with patience of 10 epochs based on validation loss prevents overfitting.

After training, the model outputs scaled log return predictions for the test period, which are inverse-transformed back to log returns. Predicted prices are reconstructed

recursively from the previous day’s actual price:

$$\hat{P}_t = P_{t-1} \exp(\hat{r}_t).$$

Models 3 & 6: LSTM and GRU with GBM Features (MSE Loss)

Models 3 (LSTM) and 6 (GRU) use the same recurrent framework but change the feature set to focus on statistical properties motivated by the GBM model. For each day, the input includes the log return, relative distance to the 20-day SMA, GBM volatility estimate over the previous 90 days, and trading volume.

These features and the log return target are standardized to have zero mean and unit variance (based on the training set) rather than Min–Max scaled. The network is slightly deeper and wider than in Models 2 and 5, with two recurrent layers of 128 units, higher dropout ($p = 0.4$), and batch normalization layers for regularization and stabilization. The loss function remains standard MSE on the standardized log returns, optimized with Adam at a learning rate of 0.0005. Early stopping with patience of 15 epochs is used.

The same reconstruction procedure is used to derive price forecasts from predicted log returns. By comparing Models 2/5 with Models 3/6, it is possible to assess whether GBM-style statistical features are more informative than traditional technical indicators for cryptocurrency price prediction.

Models 4 & 7: LSTM and GRU with Directional Penalty

Models 4 (LSTM) and 7 (GRU) share the same architecture and inputs as Models 3 and 6, respectively, but modify the training objective in an attempt to improve directional accuracy. Instead of pure MSE, the loss function upweights errors when the predicted and actual log returns have opposite signs.

Let y_t be the standardized true log return and $\hat{y}_t$ the standardized prediction for day t . The squared error is

$$e_t = (y_t - \hat{y}_t)^2.$$

If y_t and $\hat{y}_t$ have opposite signs, then $y_t \hat{y}_t < 0$ and the prediction is directionally wrong. The directional loss is defined as

$$\mathcal{L}_{\text{dir}} = \frac{1}{T} \sum_{t=1}^T \omega_t e_t,$$

with

$$\omega_t = \begin{cases} 5.0, & \text{if } y_t \hat{y}_t < 0 \text{ (wrong direction),} \\ 1.0, & \text{otherwise.} \end{cases}$$

Thus a directionally incorrect prediction counts five times as much as a directionally correct one with the same magnitude of error. The intuition is that the model will be incen-

tivized to get the sign right even if that means slightly larger magnitude errors.

All other aspects of training—optimizer, learning rate, early stopping, and reconstruction of prices—are the same as in Models 3 and 6. By comparing Models 3 with 4, and Models 6 with 7, we can evaluate whether explicitly penalizing directional mistakes improves DA in practice.

5 Results

Price Forecast Accuracy

Table 5 (see Appendix) summarizes the out-of-sample RMSE, MAE, MAPE, and DA of all seven models for the five cryptocurrencies. Lower values indicate better performance for RMSE, MAE, and MAPE; higher values indicate better performance for DA. Several patterns are clear.

First, all LSTM and GRU models substantially outperform Prophet in terms of price forecast accuracy. For Bitcoin, the RMSE falls from about \$4,091 for Prophet to around \$2,201 for the basic LSTM, and further to about \$2,162 for the GBM-enhanced variants. For XRP, MAPE drops from 8.21% for Prophet to 3.00% for the basic LSTM and below 2.9% for the GBM models. Similar gains appear across all five assets. This directly supports the conclusion that, under the chosen setup, recurrent neural networks clearly defeat Prophet in price prediction accuracy.

Second, within both the LSTM and GRU families, GBM-inspired statistical features are generally more informative than traditional technical indicators. Comparing the basic LSTM (Model 2) to the GBM-LSTM variants (Models 3 and 4), MAPE decreases for all five assets. The improvement is especially visible for XRP, where MAPE falls from 3.00% to 2.87%. Similarly, for GRU, GBM features reduce MAPE from 2.76% (average of basic GRU) to 2.49% (average of GBM-GRU-MSE). Overall, these results indicate that volatility and distance-to-trend features carry useful predictive information that complements or surpasses that in RSI and SMA.

Third, Models 3 and 4 (LSTM variants), as well as Models 6 and 7 (GRU variants), have almost identical error metrics on price levels. The directional penalty loss does not systematically change point forecast accuracy in either direction. This holds across all five assets.

LSTM vs. GRU Comparison

A natural question is whether the LSTM or GRU architecture is better suited for this task. Table 1 presents a direct comparison.

For basic models without GBM features, LSTM slightly edges out GRU on most coins (winning on 4 of 5 coins by RMSE). However, once GBM features are added, the gap narrows dramatically—the average MAPE for GBM-LSTM-MSE and GBM-GRU-MSE

Table 1: LSTM vs. GRU comparison across all coins, averaged over common feature sets and losses.

Variant	Metric	LSTM	GRU
Basic (Tech. Indicators)	Avg. MAPE	2.65%	2.76%
	Avg. RMSE	4th	5th
GBM + MSE	Avg. MAPE	2.49%	2.49%
	Avg. RMSE	2nd/3rd	2nd/3rd
GBM + Directional Loss	Avg. MAPE	2.46%	2.46%
	Avg. RMSE	1st/2nd	1st/2nd

is identical at 2.49%, and the same holds for the directional loss variants at 2.46%. This pattern strongly suggests that architecture choice (LSTM vs. GRU) matters far less than the choice of input features and, to a lesser extent, the loss function.

Directional Accuracy

Table 5 (see Appendix) also reports the directional accuracy of each model on the test set.

Across all assets and models, directional accuracies range roughly from 45% to 55%, clustering only slightly above the 50% random-guess baseline. While some models do beat 50% by a few percentage points on certain assets—for example, GBM-GRU-DL achieves 55.07% DA for BNB—the gains are modest.

For BNB, both GBM-LSTM-DL and GBM-GRU-DL achieve 55.07% directional accuracy, the highest across all models and assets. This is classified as a “Strong Directional Signal,” but it remains a single asset and should be interpreted cautiously. For most other assets, directional accuracies hover near or below 52%, indicating limited practical value for trading decisions.

Comparing features within the recurrent model family, GBM features sometimes help directional accuracy but not consistently. For BNB and Ethereum, GBM-enhanced models outperform their basic counterparts in DA. For Bitcoin and XRP the differences are minor. For Solana, the basic LSTM slightly outperforms its GBM counterpart.

Most importantly, introducing a directional penalty in the loss function (Models 4 and 7) does not systematically improve DA relative to Models 3 and 6. For Bitcoin, Model 4 falls exactly to the 50% random benchmark. For BNB, both directional loss variants maintain the 55.07% achieved by their MSE counterparts. For other assets, the differences are small and inconsistent. Taken together, these results suggest that the directional penalty, at least in the simple form implemented here (a uniform $5\times$ weight on all directionally wrong predictions), does not reliably enhance directional accuracy.

Best Model per Metric per Coin

Table 2 identifies the best model for each metric on each coin.

Table 2: Best model per metric per coin.

Coin	Best RMSE	Best MAPE	Best DA
Bitcoin	GBM-GRU-DL	GBM-LSTM/GRU-DL	GBM-GRU-MSE
Ethereum	GBM-LSTM-DL	GBM-GRU-MSE	Prophet
BNB	GBM-LSTM-DL	GBM-LSTM-DL	GBM-LSTM/GRU-DL
Solana	GBM-GRU-DL	GBM-GRU-DL	GBM-LSTM-DL
XRP	GBM-GRU-DL	GBM-GRU-DL	LSTM / GRU

The GBM-enhanced directional loss variants (GBM-LSTM-DL and GBM-GRU-DL) dominate the RMSE and MAPE categories, winning or tying for the top position on all five coins. For directional accuracy, no single model consistently outperforms, and notably Prophet achieves the highest DA on Ethereum (53.88%) despite having the worst price accuracy.

Model Ranking by Average MAPE

Table 3 ranks all seven models by their average MAPE across all five coins.

Table 3: Model ranking by average MAPE across all five cryptocurrencies.

Rank	Model	Avg. MAPE (%)
1	GBM-GRU-DL	2.46
2	GBM-LSTM-DL	2.46
3	GBM-GRU-MSE	2.49
4	GBM-LSTM-MSE	2.49
5	LSTM	2.65
6	GRU	2.76
7	Prophet	5.74

The ranking reveals a clear three-tier structure. The top tier consists of the four GBM-enhanced variants, all achieving average MAPE between 2.46% and 2.49%. The second tier contains the basic LSTM and GRU with technical indicators, at 2.65% and 2.76% respectively. Prophet sits alone in the third tier at 5.74%, more than double the error of the deep learning models.

Discussion of the Four Research Questions

The empirical findings can now be tied back to the research questions.

RQ1: LSTM/GRU vs. Prophet. LSTMs and GRUs clearly dominate Prophet in terms of point forecast accuracy. For every asset and every error metric (RMSE, MAE, MAPE), the recurrent models produce smaller errors than Prophet. This holds even for the basic models with simple technical indicators, and it becomes more pronounced for the GBM-enhanced variants. In other words, the flexibility of LSTM and GRU networks to capture nonlinear and long-range dependencies appears to be valuable when forecasting highly volatile, non-stationary cryptocurrency prices, whereas Prophet’s piecewise linear trend, which is designed mainly for structured seasonality and calendar effects, is less well matched to this setting.

RQ2: GBM features vs. technical indicators. GBM-inspired statistical features are more informative than traditional technical indicators for price prediction in this experiment. Comparing Models 2/5 to Models 3–4/6–7, we see systematic improvements in MAPE for all five assets. This suggests that measuring recent volatility and the asset’s deviation from a moving-average trend in a probabilistic way provides the recurrent model with a richer and more relevant description of the state of the market than RSI and SMA alone. From a modeling perspective, these features tie more directly to standard continuous-time asset pricing models and may better reflect the underlying dynamics of returns.

RQ3: LSTM vs. GRU. The LSTM and GRU architectures yield near-identical results when paired with the same feature set and loss function. With GBM features, the average MAPE for LSTM and GRU variants is effectively the same (2.46% for directional loss, 2.49% for MSE loss). The basic LSTM has a slight edge over the basic GRU (2.65% vs. 2.76% average MAPE), but this advantage disappears once GBM features are introduced. This finding is consistent with the broader machine learning literature suggesting that LSTM and GRU are roughly equivalent in expressive power for most sequence modeling tasks, and that feature engineering often matters more than the choice of recurrent cell.

RQ4: Directional penalty loss. The directional penalty does not deliver the hoped-for improvement in directional accuracy. Model 4 and Model 7 were designed to care more about getting the sign of the return correct, even at the expense of slightly larger magnitude errors, but the realized DA remains very close to that of Models 3 and 6 and sometimes worse. A plausible interpretation is that for these data, there is limited exploitable structure in short-term direction beyond what is already captured by minimizing squared error. Penalizing misdirection more heavily might simply encourage the model to make more conservative predictions near zero, which marginally changes the sign pattern but does not generate robust directional skill.

At a higher level, the directional accuracy numbers reinforce a common finding in empirical finance: one-day-ahead price direction in highly liquid markets is extremely hard to predict, and improvements over a 50% baseline are small and fragile. The fact that all

models achieve low price errors but only modest directional gains suggests that reducing magnitude error does not automatically translate into directional predictability.

6 Conclusion

This project investigated next-day closing price prediction for five major cryptocurrencies, Bitcoin, Ethereum, BNB, Solana, and XRP, using both a traditional time-series model (Facebook Prophet) and several deep learning models implemented in PyTorch.

Under the specific experimental conditions (daily OHLCV data from Bitget, a 90-day lookback window, and a 70%/15%/15% train/validation/test split), the results can be summarized as follows.

First, LSTM and GRU models with appropriate feature engineering substantially reduce price forecast errors relative to Prophet for all five cryptocurrencies. RMSE, MAE, and MAPE all improve, sometimes by large margins, indicating that flexible sequence models can better accommodate the noisy and nonlinear dynamics of cryptocurrency prices than a simple trend-only Prophet specification.

Second, within the recurrent model family, GBM-inspired statistical features lead to the best overall price accuracy. These features, volatility and distance-to-trend, appear to condense relevant information about the recent distribution of returns more effectively than standard technical indicators like RSI and SMA, especially for assets such as XRP.

Third, LSTM and GRU produce nearly indistinguishable results when using the same features and loss function. This indicates that the choice of recurrent architecture is secondary to the choice of input features and loss function for this prediction task.

Fourth, explicitly penalizing directionally wrong predictions in the loss function does not provide a robust improvement in directional accuracy. The directional loss variants achieve similar DA to their MSE counterparts, and in some cases slightly worse results. This suggests that, at least with the simple implementation used here, adjusting the loss function is not sufficient to overcome the fundamental difficulty of short-term direction prediction in volatile markets.

Two caveats are important. Neural network training is inherently stochastic, and the reported numbers are based on a particular random seed and data split. To assess the robustness of these conclusions, further work should repeat the experiments over multiple random seeds, alternative train/test splits, and possibly across different exchanges. In addition, the evaluation focuses on statistical forecasting error rather than economic performance; incorporating transaction costs, execution delays, and position sizing would be necessary to translate predictive models into trading strategies.

Looking ahead, the most promising direction may not be to chase a single “correct” point prediction or a slightly higher directional accuracy, but rather to develop models that provide reliable probabilistic forecasts and explicit measures of uncertainty. Such models could help market participants better assess risk, stress-test portfolios under different sce-

narios, and make more informed decisions in an environment where, as the Latin phrase reminds us, *scientia potentia est*—knowledge is power.

Appendix

Key Hyperparameters

Table 4 summarizes the most important configuration choices across models.

Component	Setting
Lookback window (all recurrent)	$W = 90$ days
Prophet training window	Last 90 days (rolling per test point)
Train/val/test split (recurrent)	70% / 15% / 15% (chronological)
Prophet test ratio	15% (chronological)
Target (recurrent)	Next-day log return r_{t+1}
Target (Prophet)	Next-day closing price P_{t+1}
Models 2 & 5 (LSTM/GRU with Technical Indicators)	
Hidden size	64
Num. layers	2
Dropout	0.2
Batch normalization	No
Optimizer	Adam, learning rate 0.001
Loss	Mean Squared Error (MSE)
Feature scaling	Min-Max (0, 1)
Max epochs	50
Early stopping patience	10
Models 3, 4, 6, 7 (LSTM/GRU with GBM Features)	
Hidden size	128
Num. layers	2
Dropout	0.4
Batch normalization	Yes
Optimizer	Adam, learning rate 0.0005
Loss (MSE variants)	Mean Squared Error (MSE)
Loss (Dir. variants)	Directionally-weighted MSE (5× penalty)
Feature scaling	StandardScaler (zero mean, unit variance)
Max epochs	50
Early stopping patience	15
Common	
Batch size (all recurrent)	32
Prophet seasonality	Daily = weekly = yearly = disabled
Prophet changepoint prior scale	0.05

Table 4: Key hyperparameters and configuration choices for all models.

Detailed Results

Asset	Model	RMSE	MAE	MAPE (%)	DA (%)
Bitcoin	Prophet	4090.91	3301.78	3.33	48.62
	LSTM	2201.29	1635.93	1.62	50.81
	GBM-LSTM-MSE	2198.73	1613.91	1.60	50.00
	GBM-LSTM-DL	2162.05	1575.96	1.57	50.00
	GRU	2272.03	1720.76	1.70	50.81
	GBM-GRU-MSE	2188.63	1605.20	1.59	52.03
	GBM-GRU-DL	2161.82	1575.62	1.57	50.00
Ethereum	Prophet	229.48	180.29	5.98	53.88
	LSTM	122.56	84.35	2.80	49.19
	GBM-LSTM-MSE	124.95	85.87	2.82	47.64
	GBM-LSTM-DL	122.64	83.32	2.77	50.68
	GRU	124.12	86.19	2.84	49.51
	GBM-GRU-MSE	122.67	83.78	2.77	52.03
	GBM-GRU-DL	122.88	83.65	2.78	50.68
BNB	Prophet	42.08	29.24	3.73	52.38
	LSTM	27.77	18.86	2.32	45.60
	GBM-LSTM-MSE	26.15	16.60	2.01	45.27
	GBM-LSTM-DL	25.94	16.20	1.96	55.07
	GRU	29.36	20.51	2.52	45.60
	GBM-GRU-MSE	26.82	17.37	2.10	44.93
	GBM-GRU-DL	25.98	16.26	1.97	55.07
Solana	Prophet	15.19	12.36	7.44	47.57
	LSTM	8.28	6.35	3.50	49.77
	GBM-LSTM-MSE	7.42	5.65	3.14	50.49
	GBM-LSTM-DL	7.30	5.58	3.13	51.46
	GRU	8.02	6.12	3.39	49.77
	GBM-GRU-MSE	7.40	5.66	3.14	50.00
	GBM-GRU-DL	7.29	5.57	3.12	50.00
XRP	Prophet	0.25	0.19	8.21	53.63
	LSTM	0.11	0.08	3.00	51.47
	GBM-LSTM-MSE	0.11	0.07	2.87	52.70
	GBM-LSTM-DL	0.11	0.07	2.88	48.31
	GRU	0.12	0.09	3.33	51.47
	GBM-GRU-MSE	0.11	0.07	2.86	50.34
	GBM-GRU-DL	0.11	0.07	2.86	51.69

Table 5: Test set results for all models. RMSE and MAE are in USD; MAPE and DA are in percent.

Visualization for Bitcoin

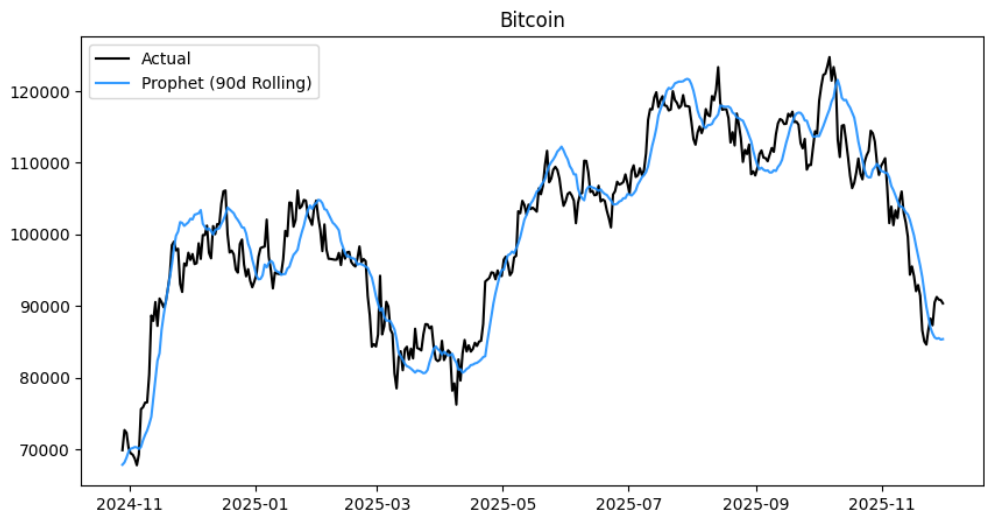


Figure 1: Prophet (Model 1) – Bitcoin Next-Day Closing Price Prediction.

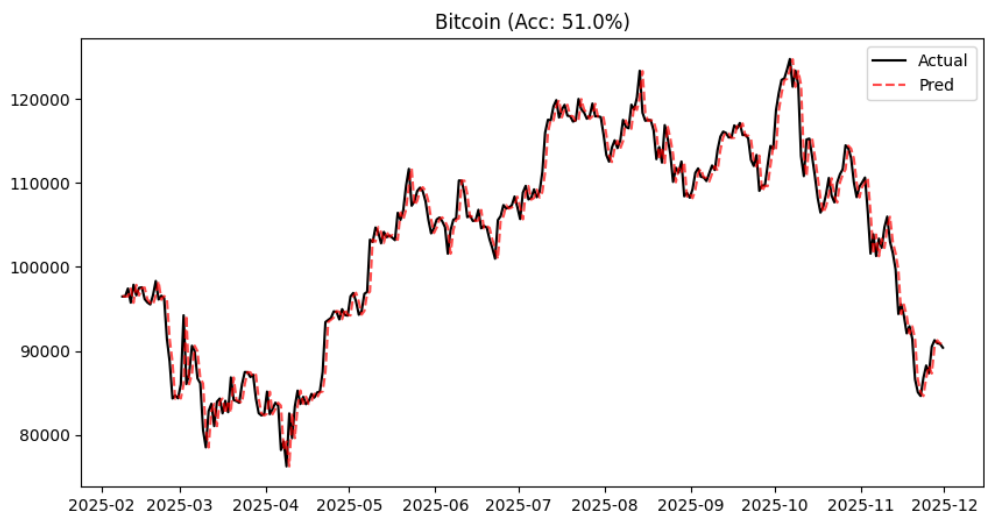


Figure 2: LSTM (Model 2) – Bitcoin Next-Day Closing Price Prediction (with directional accuracy).